

Sai Rahul Bhat Gaadhi Raju

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Career Objective

B.Tech Electronics and Communication Engineering student passionate about VLSI Design, Semiconductor Technology, RTL Design, and Digital Systems. Selected for Samsung ISWDP Cohort 8 Fellowship and actively building expertise in semiconductor device simulation, digital design, and chip development. Seeking opportunities in VLSI, semiconductor engineering, and hardware design.

Education

B.Tech in Electronics and Communication Engineering	2024 – Till Date
Anurag University, Hyderabad CGPA: 8.01	
Diploma in Electronics and Communication Engineering	2021 – 2024
JNGPT, Ramanthapur CGPA: 9.17	

Achievements

- Selected for Samsung Fellowship – ISWDP Cohort 8 (Grade II) among 4000+ applicants
- Selected for India Semiconductor Workforce Development Program (ISWDP) focused on Semiconductor Technology and Synopsys TCAD Training

Technical Skills

HDL Programming: Verilog HDL, Python, C (Basics)
VLSI Semiconductor: RTL Design, Digital Electronics, CMOS Fundamentals, Semiconductor Devices, ASIC Design Flow, Static Timing Analysis (STA), Physical Design Fundamentals
Networking: TCP/IP, Routing, Switching, VLAN Fundamentals, Network Security Basics
Tools: Vivado, Cadence Virtuoso, Synopsys Sentaurus TCAD (Training), MATLAB, Cisco Packet Tracer
Coursework: Digital Electronics, Analog Electronics, Signals Systems, Semiconductor Devices, VLSI Design

Projects

Pipelined ALU Design (RTL – Verilog)

- Designed multi-stage pipelined ALU based on RISC-V architecture to improve throughput
- Developed pipeline stages considering timing constraints and hardware efficiency
- Developed and verified RTL using Verilog HDL and Vivado simulation environment

Barrel Shifter Design (Verilog HDL)

- Designed N-bit barrel shifter using combinational logic techniques
- Implemented efficient bit shifting through RTL modeling
- Verified functionality using simulation tools

4-bit ALU Design (Verilog HDL)

- Designed arithmetic and logical operations using modular RTL architecture
- Performed simulation and verification using Vivado

VLSI Voyage: 16-bit ALU (Pipeline vs Non-Pipeline)

- Compared pipelined and non-pipelined ALU architectures
- Analyzed throughput, latency, timing performance, and clock improvements
- Evaluated trade-offs between performance and hardware complexity

Certifications

- NPTEL – VLSI Physical Design with Timing Analysis (IIT Roorkee)
- NPTEL – Digital Design with Verilog (IIT Guwahati)
- NPTEL – VLSI Design Flow: RTL to GDS (IIIT Delhi)
- Maven Silicon – VLSI SoC Design Overview

Internship

Python Intern – Vault of Codes

1 Month

- Worked on Python fundamentals and logic-based problem solving
- Developed basic automation and programming skills through practical assignments

Languages

English, Hindi, Telugu